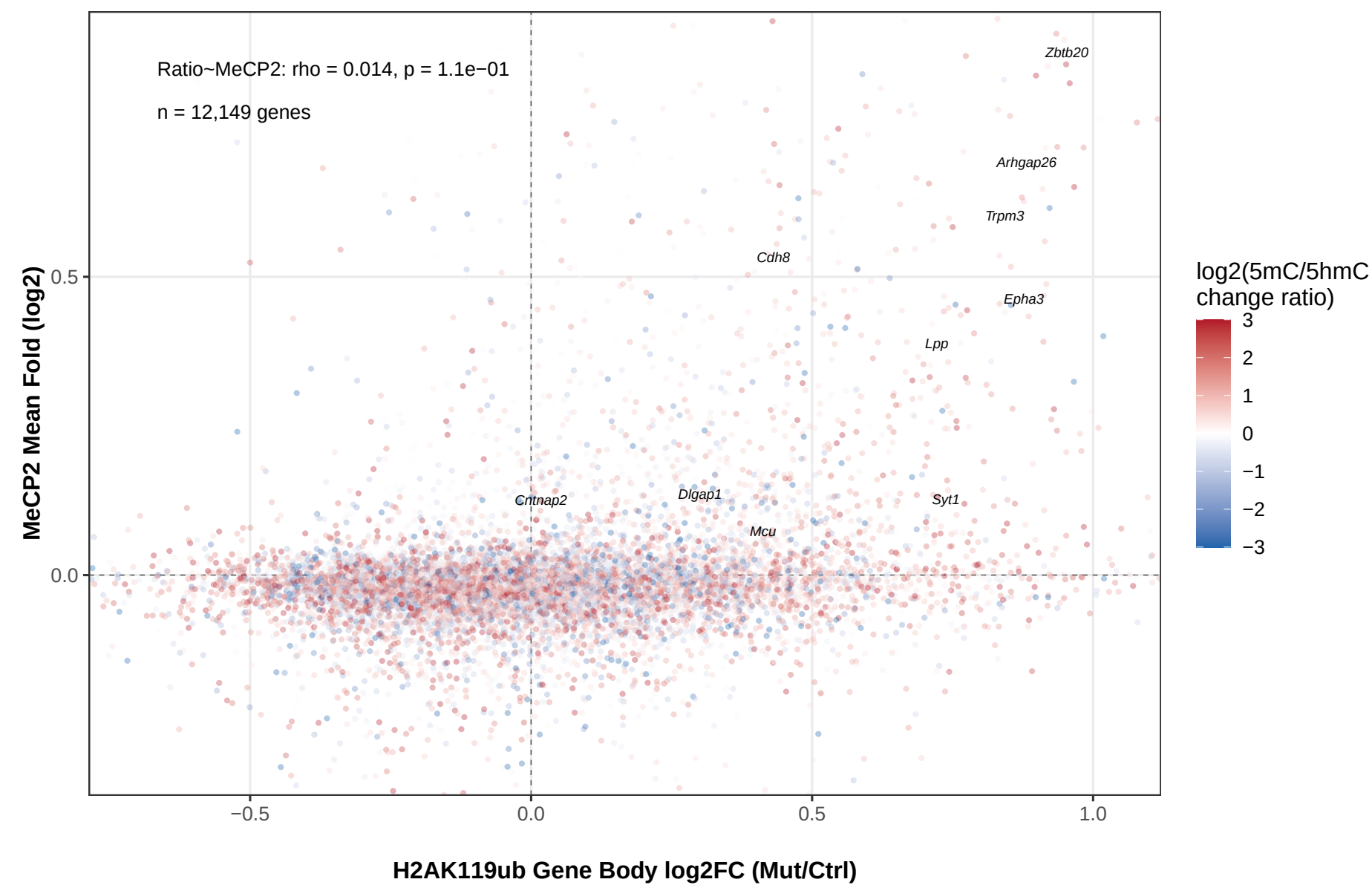


5mC/5hmC Ratio vs MeCP2 at H2AK119ub Loci

Does MeCP2 binding scale with the degree of 5mC-skew (gradient), or respond equally above ratio=1 (threshold)?

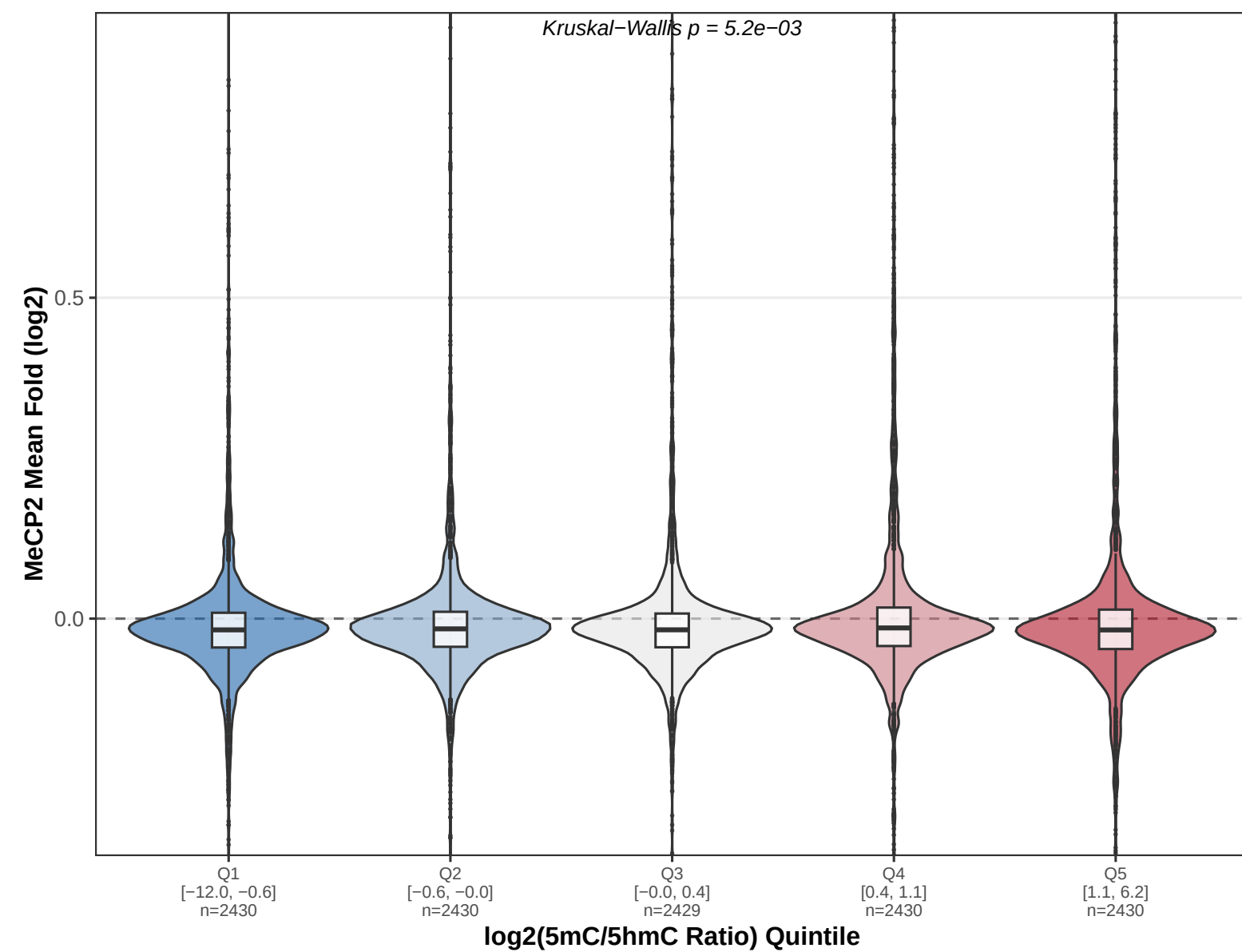
MeCP2 Binding vs H2AK119ub, Colored by 5mC/5hmC Ratio

Red = mC gain dominates | Blue = hmC loss dominates | White = stoichiometric



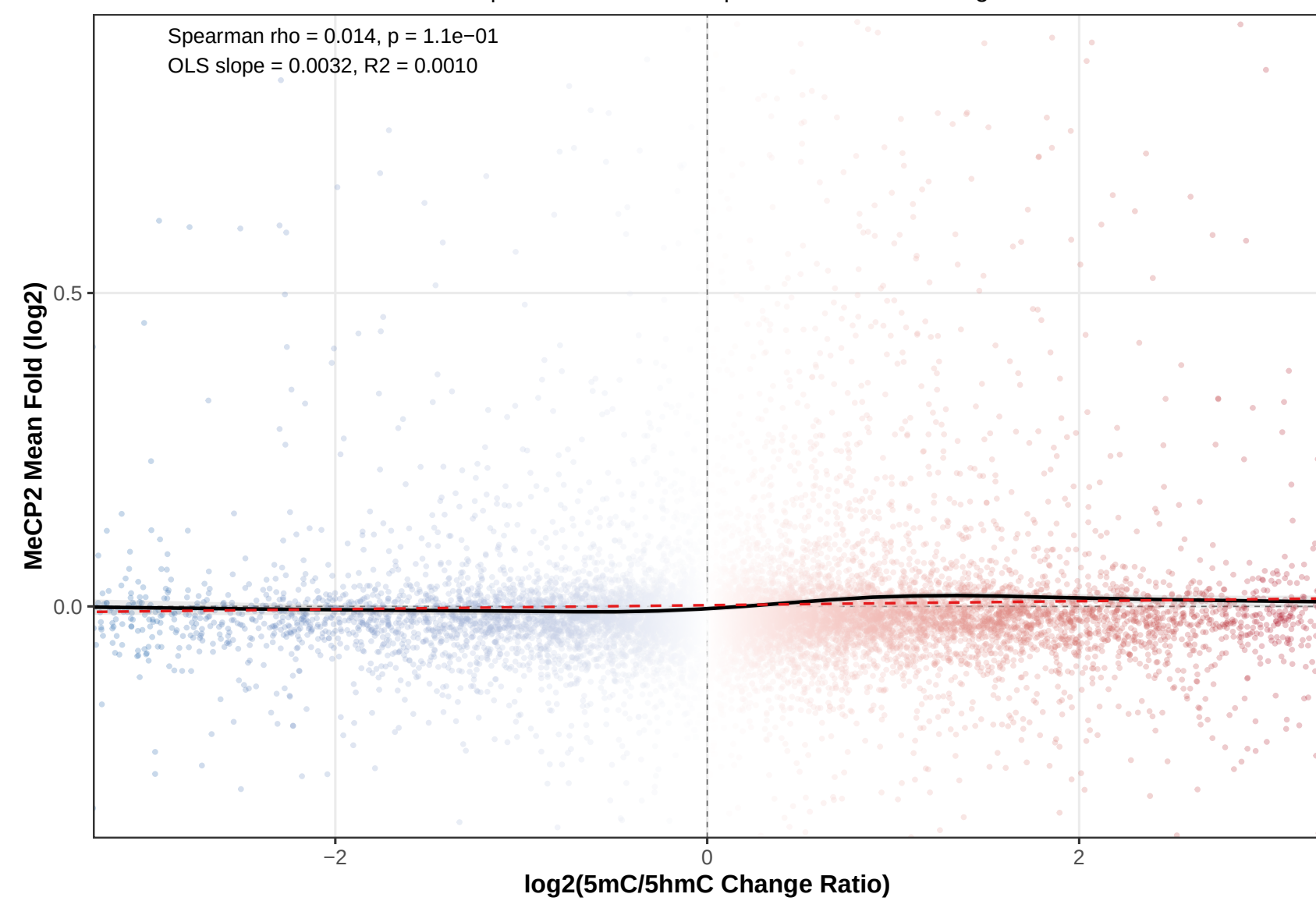
MeCP2 Fold by 5mC/5hmC Ratio Quintile

Gradient: MeCP2 increases across quintiles | Threshold: flat above Q3



5mC/5hmC Ratio vs MeCP2 Fold (Direct)

Black = loess | Red dashed = OLS | Is there a continuous gradient?



Nested Model Comparison: Predicting MeCP2 Fold

Interaction (K119ub x Ratio): est=+0.0204, $p=4.30e-14$ | $n=12,149$

